YouTube Walkthrough Scripts (with exact cells)

How to find a cell quickly in Colab: click the **Table of contents** icon (☰) in the left sidebar. Every heading listed below appears there, so click a heading to jump to it. "Cell #" counts every cell (text and code) from the top of the notebook.

Before recording: `Runtime → Run all` so all outputs are visible. Have the GitHub repo open in a second tab and the Streamlit app running in a third.

Each video is laid out as a table: - **Go to** is the heading and cell number to scroll to. - **Point at** is what to highlight with your cursor. - **Say** is your narration.

Video 1: CRISP-DM | Telco Customer Churn (≈8–10 min)

#	Go to	Point at	Say
1	GitHub repo README	the 3-project table	"Hi, I'm Anshita Shah, MSSE at San José State. This is the first of three videos for my CMPE 255 data mining process assignment: a telecom churn project taken through all six phases of CRISP-DM."
2	**Cell 1**: title	the phase/task table	"The notebook follows the CRISP-DM 1.0 guide, and every phase covers all its generic tasks. Data Understanding, for example, has collect, describe, explore and verify quality."
3	**Cell 3**: *Phase 1 — Business Understanding*	the business success criteria and the cost assumptions list	"With 26.5% churn, a model that says nobody churns is 73.5% accurate and useless. So success is defined in business terms. I set a cost model: a $100 offer, a 35% save rate, and a saved customer worth 12 months of charges. The data-mining criteria are AUC of at least 0.80 and recall of at least 0.70."
4	**Cell 4**: code	`OFFER_COST`, `SAVE_RATE`, `expected_profit`	"These assumptions are parameters in code, and this profit function is used later to choose the threshold."
5	**Cells 5–6**: *2.1 Collect initial data*	output "rows: 7,043 columns: 21"	"The IBM Telco dataset from Kaggle: 7,043 customers, 21 attributes."
6	**Cells 7–9**: *2.2 Describe data*	the attribute groups printout	"I grouped the attributes into demographics, account, services and the target."
7	**Cell 12**: *2.3 Explore data*	the churn-rate-by-category bar charts (Contract panel)	"Month-to-month contracts churn far more, and so do fiber customers, electronic-check payers, and customers without tech support."
8	**Cell 13**	Cramér's V table: Contract 0.41 at top, gender 0.008 at bottom	"Statistically, contract is the strongest signal at 0.41. Gender is essentially zero."
9	**Cell 16**: *2.4 Verify data quality*	"TotalCharges non-numeric: 11" and "tenure of those blank rows: [0]"	"TotalCharges has 11 blanks. Instead of mean-imputing, I checked why. All of them have tenure zero: new customers who haven't been billed yet. So the correct fill is zero."
10	**Cells 18–21**: *3.1 Select / 3.2 Clean*	the `drop(columns=…)` line	"I drop the ID, gender and phone service, convert TotalCharges, and collapse the 'No internet service' levels."

#	Go to	Point at	Say
11	**Cells 22–23**: *3.3 Construct data*	the feature rationale table	"Six engineered features, each with a business reason: add-on count, recent price increase, tenure bucket, auto-pay and family."
12	**Cell 25**: *3.5 Format data*	train/validation/test counts and the `ColumnTransformer`	"A stratified 60/20/20 split, with encoding inside a pipeline so nothing leaks from test into training."
13	**Cells 26–27**: *Phase 4 — Modeling*	technique table, then the CV results table	"Five candidates compared with 5-fold CV on training only. Logistic regression, random forest and gradient boosting are within about one standard deviation, around 0.85 AUC, so AUC can't pick the winner."
14	**Cell 29**: *4.3 Build model*	the three "best CV AUC … params" lines	"I tuned the three strongest families with grid search."
15	**Cell 31**: *4.4 Assess model*	ROC and PR curves	"On validation they're practically tied."
16	**Cell 33**: *Phase 5 — Evaluation*	profit curve; the dotted "Offer to everyone" line; the legend maxima	"This is the key chart: expected net value at every threshold. Offering to everyone loses money. Random forest peaks at about $43k at threshold 0.56, slightly ahead of logistic regression even though logistic regression had the higher AUC. That's why CRISP-DM evaluates on business criteria."
17	**Cell 35**: *Final, one-time test-set evaluation*	"Test ROC-AUC 0.8468 PASS", "recall 0.741 PASS", "Net value — model $37,497" vs "everyone $-26,597", confusion matrix	"On the untouched test set: AUC 0.847, recall 0.74, and +$37.5k net value versus −$26.6k for offering to everyone. Both criteria pass."
18	**Cell 36**	gains chart annotation "top 20% → ~51% of churners"	"Calling the riskiest 20% of customers catches about half of all churners."
19	**Cell 38**: *Explainability*	importance bars, Contract at top	"The top drivers are contract, internet service and tenure. The retention team can act on all three."
20	**Cell 39**: *5.2 Review process*	the checklist table	"I reviewed the process for leakage and test discipline, and noted the gaps: no temporal validation and no uplift model."
21	**Cell 41**: *Phase 6 — Deployment*	`prepare_features`, the `assert`, `joblib.dump`, printed test metrics	"Deployment saves one artifact with the pipeline, threshold and feature list. The assert proves the scoring code reproduces the training features."

#	Go to	Point at	Say
22	**Cell 42**	the 5-customer table with OFFER actions	"Here's a smoke test on unseen customers."
23	Terminal	run `python app/score_batch.py data/Telco-Customer-Churn.csv call_list.csv`	"The batch scorer turns a CSV into a ranked call list."
24	Streamlit app	change Contract to "Two year" and watch the risk drop	"And this is the retention-desk app."
25	**Cell 44**: *6.2 Monitoring*	PSI table: 0.015 vs 1.399 → RETRAIN	"Monitoring uses PSI. Train to test is stable at about 0.01. A simulated 15% price hike gives 1.4, which triggers a retrain alert."
26	`critic_logs/critic_review.md` on GitHub	a Round 1 MAJOR item	"Each phase was critiqued by an LLM playing a CRISP-DM authority. It caught leakage and a missing cost model in my drafts. Links are in the description. Thanks!"

Video 2: SEMMA | Medical Insurance Cost (≈6–8 min)

#	Go to	Point at	Say
1	**Cell 1**: title	the SEMMA ↔ SAS node table	"Video two uses SAS's SEMMA: Sample, Explore, Modify, Model, Assess. It has no business or deployment phase. I mapped each step to its SAS Enterprise Miner node, and the repo has a SAS program for SAS OnDemand."
2	**Cell 4**: *S.1 Input data*	roles table	"Six inputs, and charges as the target."
3	**Cells 6–8**: *S.2 Data integrity*	the two duplicate rows, then "Rows after de-duplication: 1337"	"One exact duplicate. I removed it before partitioning so it can't land in both train and test."
4	**Cell 10**: *S.3 Sampling strategy*	partition profile table (smoker % ≈ 20.5 everywhere)	"With 1,337 rows I use all the data. The sampling decision is a 60/20/20 partition stratified on smoker and charge quartile."
5	**Cell 11**	"All p-values > 0.05 … True" and the p-value table	"KS and chi-square tests confirm every partition matches training."
6	**Cells 13–14**: *E.1 Univariate*	"Skewness of charges: 1.54", the charges vs log(charges) histograms	"Charges are right-skewed. The log looks symmetric. That's a hypothesis I test later."
7	**Cell 16**: *E.2 Bivariate*	right-hand scatter (BMI vs charges, jump at 30 for smokers)	"Smokers jump in cost at BMI 30. Non-smokers barely change."
8	**Cell 17**	pivot: 21,651 vs 42,035	"Smokers with BMI over 30 average about $42k, double other smokers. That's the key interaction."
9	**Cell 20**: *E.3 Outlier analysis*	"smokers among them: 97%"	"The outliers are 97% smokers, so they're real cases and I keep them."
10	**Cells 22–23**: *M — Modify*	the transformation table, then `modify()`	"Every new variable traces back to an Explore finding: obese, smoker×obese, smoker×BMI, age²."
11	**Cell 27**: *M.3 Variable selection*	mutual information list	"Mutual information confirms age and the smoker interactions matter most."
12	**Cell 30**: *M — Model*	the six `models[...]` lines and best-params output	"Six candidates: raw regression, engineered regression in dollars and in log, a tree, a forest and boosting."

#	Go to	Point at	Say
13	**Cell 33**: *A.1 Model comparison*	table with "Regression (modified)" at top (4,279 / 0.860) and "modified, log" at bottom (0.453)	"The surprise: the engineered linear regression wins, beating both ensembles. Modify lifted regression from R^2 0.73 to 0.86. The log target was worst at 0.45, because the smoker effect is a dollar surcharge. That's why SEMMA separates Explore from Assess."
14	**Cell 36**: *A.2 Champion on test*	printed scores (R^2 0.875) and the predicted-vs-actual plot	"On the untouched test set: R^2 0.875, RMSE $4,261."
15	**Cell 38**: *A.3 Decile assessment*	decile table, last row "3.13"	"The model ranks risk correctly. The top decile is 3.1 times the average."
16	**Cell 41**	coefficients: smoker_bmi +5,519, smoker_obese +5,008	"And it's interpretable: each coefficient is a dollar effect."
17	**Cell 43**	the three score examples ($6,551 / $19,449 / $39,533)	"Same person: non-smoker, smoker, and obese smoker."
18	Streamlit app	toggle Smoker yes/no, move BMI past 30	"SEMMA stops at Assess. This app just demonstrates the hand-off."
19	`sas/semma_insurance.sas` on GitHub	the `PROC SURVEYSELECT` and `PROC GLMSELECT` blocks	"The SAS program reproduces the same flow in SAS OnDemand. Thanks!"

Video 3: KDD | Grocery Market Basket (≈6–8 min)

#	Go to	Point at	Say
1	**Cell 1**: title	the KDD flow diagram and step table	"The last video applies Fayyad's KDD process to almost 10,000 grocery baskets. Data mining is only one of its steps."
2	**Cell 3**: *Step 0*	success criteria bullet	"Knowledge must be valid, novel, useful and understandable, and I gave each a measurable test."
3	**Cell 5**: *Step 1 — Selection*	"Transactions selected: 9,835" and sample baskets	"Each line is a basket. I select all of them, with item presence."
4	**Cells 7–8**: *Step 2 — Preprocessing*	top-items list ("shopping bags" visible), basket-size and support charts	"Whole milk is in a quarter of baskets. Notice 'shopping bags' in the top ten. It's not a product."
5	**Cells 9–10**	preprocessing decisions, then "Baskets after preprocessing: 9,787"	"So I remove it, which drops 48 bag-only baskets. I keep single-item baskets to avoid inflating support."
6	**Cell 12**: *Step 3 — Transformation*	"Items kept after reduction: 119 of 169", discovery/hold-out shapes	"Transformation: a Boolean matrix, reduction to 119 items, and 20% of baskets held out to validate later."
7	**Cell 13**	department counts	"And a concept hierarchy from items to departments for multi-level mining."
8	**Cell 15**: *Step 4 — Data Mining*	"Apriori: 322 … FP-Growth: 322 … identical: True"	"Apriori and FP-Growth give identical itemsets, which checks correctness."
9	**Cell 16**	runtime chart	"Runtime grows as support drops."
10	**Cells 17–18**	"Item-level rules with lift > 1: 592", department-pair table	"592 item rules and 44 department-pair rules."
11	**Cell 20**: *5.1 Interestingness*	scatter coloured by lift	"Confidence is biased toward staples, so I use lift, leverage and conviction."
12	**Cell 22**: *5.2 Statistical validity*	"significant rules: 468 of 592"	"A Bonferroni-corrected Fisher test keeps 468."
13	**Cell 24**: *5.3 Redundancy*	"468 → 466"	"Redundancy pruning keeps 466."

#	Go to	Point at	Say
14	**Cell 26**: *5.4 Hold-out validation*	"454 of 466 (97%)", hold-out lift plot	"454 of them, 97%, still have lift above one on baskets the algorithm never saw. That's validity on new data."
15	**Cells 27–28**: *KDD iteration*	"Final knowledge base: 112 rules (18 …)", top novel rules: {beef} → {root vegetables} 3.05 / 2.92	"The first pass was full of rules ending in whole milk: valid but obvious. So I looped back and tightened the criteria. The final base has 112 rules, 18 of them novel. Beef leads to root vegetables at lift 3, holding at 2.9 on hold-out: a meal-preparation mission."
16	**Cell 29**	lift heatmap	"Pairwise lift among the top items."
17	**Cell 30**: *5.5 Knowledge consolidation*	the pattern → action table	"Here's the knowledge translated into store actions."
18	**Cell 33**: *Using the discovered knowledge*	"Rule 14.9% / Popularity 27.2% / Hybrid 27.6%"	"Usefulness test: alone, the rules lose to recommending popular items. They're precise but sparse. A hybrid reaches 27.6%. I report this honestly. The knowledge is best for merchandising and re-ranking."
19	Streamlit app	select "beef" and "root vegetables"	"The app explains why each item is recommended. Thanks for watching all three. Links are below!"